

Passive Reconnaissance

Research Report

Introduction

What is Passive reconnaissance?

Passive reconnaissance, often shortened to passive recon, is the essential initial step in a cybersecurity assessment or bug bounty program where a researcher systematically gathers intelligence about a target without directly interacting with or probing its live network services. This non-intrusive phase relies exclusively on publicly available information and third-party sources, such as search engines, WHOIS databases, social media, and historical website archives like the Wayback Machine, ensuring a minimal risk of detection by the target's security systems. The primary objective is to build a comprehensive, low-profile map of the target's attack surface, uncovering hidden assets, technology stacks, exposed credentials, and organizational details, which then guides and focuses all subsequent, more detectable active testing efforts.

What can we gather from Passive reconnaissance?

1. Organizational information

Company name, subsidiaries, business structure, physical locations, contact details, and publicly available documents (reports, press releases, job postings).

2. Domain and network details

Registered domains and subdomains, domain ownership data (WHOIS), name servers, IP address ranges, autonomous system numbers (ASNs), and hosting providers.

3. Technology stack information

Web technologies in use such as server software, frameworks, content management systems, programming languages, analytics tools, and third-party services, inferred from public sources.

4. Digital footprint and online presence

Official websites, blogs, social media accounts, developer platforms, forums, and code repositories that reference the organization or its assets.

5. Employee and user information

Employee names, job roles, email address patterns, and organizational hierarchy obtained from social networks, professional profiles, and public disclosures.

6. Security posture indicators

Publicly disclosed vulnerabilities, past security incidents, exposed credentials in data breaches, SSL/TLS certificate details, and security-related headers visible in public responses.

7. Historical and contextual data

Archived versions of websites, DNS history, previous IP allocations, and changes in infrastructure over time.

Tools used for passive reconnaissance

There are several tools used for passive recon including search engines like shoudan.

Wappalyzer

See what stack the sites are running on.

WHOIS Footprinting

Which tells who's owning the website.

DNS Footprinting - Using commands such as **dig** and tools such as **dnsrecon** and **dnsdumpster.com**

Web Footprinting

Using Burp Suite, ZA Proxy, Paros Proxy, Website Informer and Firebug.

Network Footprinting

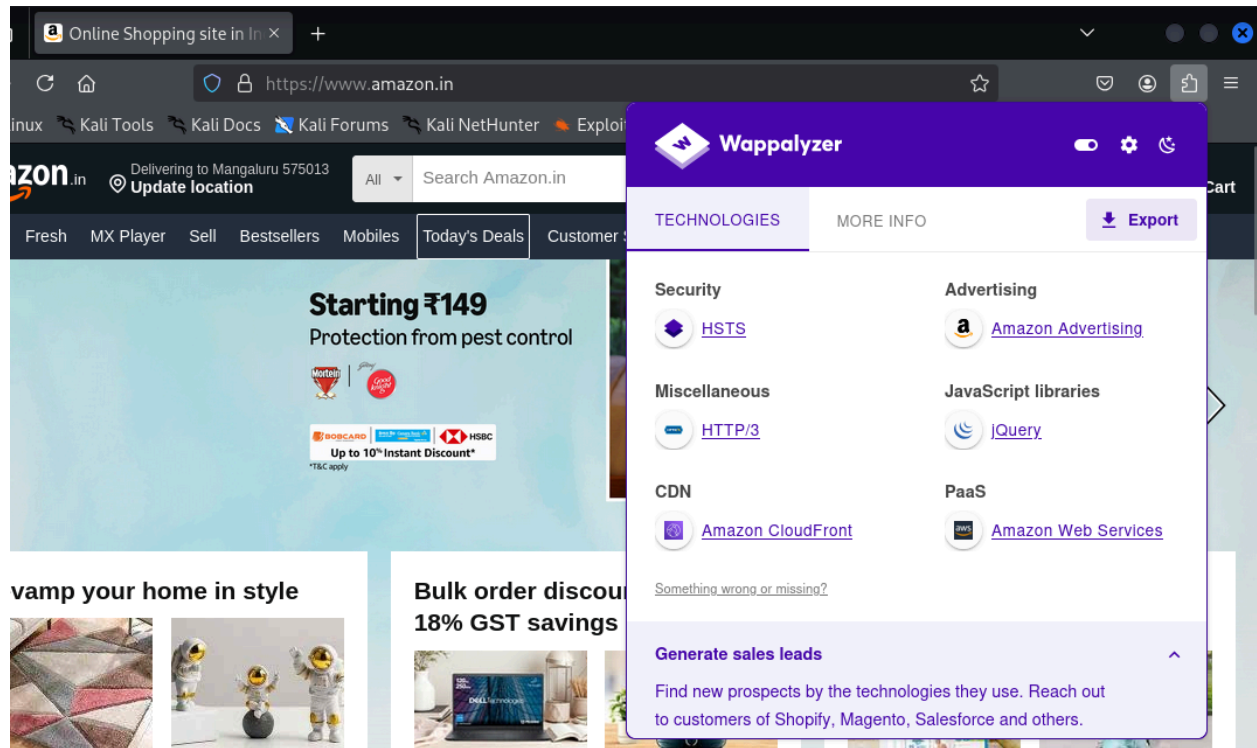
Using **nmap**, wireshark, cisco packet tracer
Tcpdump, windump and recon-ng.
Netsparker, Metasploit and Ettercap.

Misc

One can search for the target's job description, their github, linkedin, instagram, facebook and all other social media.

Practical

Using **Wappalyzer** to see which frameworks the website uses.



The screenshot shows the Wappalyzer web application interface. The left pane displays the Amazon India homepage with a banner for 'Starting ₹149 Protection from pest control' and product listings. The right pane, titled 'TECHNOLOGIES', lists detected technologies categorized into Security, Advertising, Miscellaneous, CDN, and PaaS. The 'Export' button is visible in the top right of the right pane.

Category	Technology
Security	HSTS
Advertising	Amazon Advertising
Miscellaneous	HTTP/3
CDN	Amazon CloudFront
PaaS	Amazon Web Services

Additional features shown include 'MORE INFO', 'Generate sales leads', and a link to 'Something wrong or missing?'.

Using **dig** command

```
(kali@kali)-[~]
$ dig amazon.in

; <<>> DiG 9.20.11-4-Debian <<>> amazon.in
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 36281
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:;; udp: 1232
;; QUESTION SECTION:
;amazon.in.                IN      A

;; ANSWER SECTION:
amazon.in.                 782     IN      A      3.253.168.43
amazon.in.                 782     IN      A      3.253.176.101
amazon.in.                 782     IN      A      3.253.170.100

;; Query time: 47 msec
;; SERVER: 192.168.122.1#53(192.168.122.1) (UDP)
;; WHEN: Sun Dec 14 22:17:48 IST 2025
;; MSG SIZE rcvd: 86

(kali@kali)-[~]
```

Using **nmap** to scan the site

```
(kali㉿kali)-[~]  
$ nmap amazon.in  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-14 22:20 IST  
Nmap scan report for amazon.in (3.253.168.43)  
Host is up (0.067s latency).  
Other addresses for amazon.in (not scanned): 3.253.176.101 3.253.170.100 64:ff9b:  
fd:aa64 64:ff9b::3fd:a82b  
rDNS record for 3.253.168.43: ec2-3-253-168-43.eu-west-1.compute.amazonaws.com  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https
```

Conclusion

Passive reconnaissance is a critical foundation for any effective cybersecurity assessment, penetration test, or bug bounty engagement. By systematically collecting intelligence from publicly available and third-party sources, it enables researchers to develop a clear understanding of a target's organizational structure, digital footprint, technology stack, and potential security weaknesses without alerting the target or introducing risk. The information gathered during this phase helps identify hidden assets, understand infrastructure relationships, and highlight areas of interest that may warrant deeper investigation.

Moreover, the use of specialized tools and OSINT techniques significantly enhances the accuracy and efficiency of passive recon efforts. When performed thoroughly, passive reconnaissance reduces unnecessary noise in later stages, improves the precision of active testing, and increases the likelihood of discovering meaningful vulnerabilities. Overall, passive reconnaissance is not merely a preparatory step, but a strategic process that shapes the direction, scope, and success of the entire security assessment.